

Bidirectional Mass Spectrometry Modeling Aided by Auxiliary Intermediate Steps From Graph Transformation Tools

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Institut für Theoretische Chemie
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Supervisor: ao. Univ.-Prof. Mag. Dr. Christoph Flamm

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MS-Thesis

Bidirectional Mass Spectrometry Modeling
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Hi everyone. I'm Philipp, and today I'd like to show you my master's thesis project. I'm doing it at the University of Vienna, in the TBI group.

Outline

- 1 Problem
- 2 Background
- 3 Data
- 4 ML-Modelling
- 5 Results
- 6 Conclusions

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└ Outline

Just a quick roadmap, so you know where we're going. I'll start with the problem itself and a bit of background on mass spectrometry. Then the data, and how I generate fragment candidates with MØD. After that comes the machine-learning model and the results. And at the end I'll say what I take away from all of it.

Outline

- Problem
- Background
- Data
- ML-Modelling
- Results
- Conclusions

What is my thesis about?

M ... Molecular structure S ... Mass spectrum

- Formulated mathematically as a function:

$$f : M \rightarrow S$$

- Investigation of the **inverse problem**:

$$f^{-1} : S \rightarrow M$$

- Integration of **chemically studied reaction rules** modeled in **MØD**
- Use of **derivation graphs** (rule-level fragmentation relations) to support machine-learning-based MS models

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└ Problem

└ What is my thesis about?

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So what is this project actually about?

The starting point is quite simple. I treat mass-spectrometry modelling as a function: you put in a molecular structure, and you get out a mass spectrum. That's the forward direction.

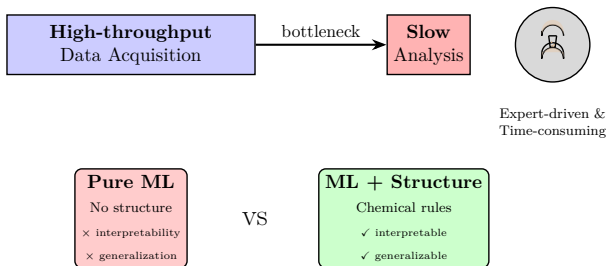
What I'm really after, though, is the inverse problem — you have a spectrum, and you want to get back to the structure.

And the idea I'm exploring is to bring the relationships between the fragments into that.

I model those with MØD, and then use them to help the machine-learning models along.

Motivation

MS is widely used, but analysis is the challenge



Acquisition fast, interpretation slow $\Rightarrow$ the bottleneck.

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Background

Motivation

Motivation



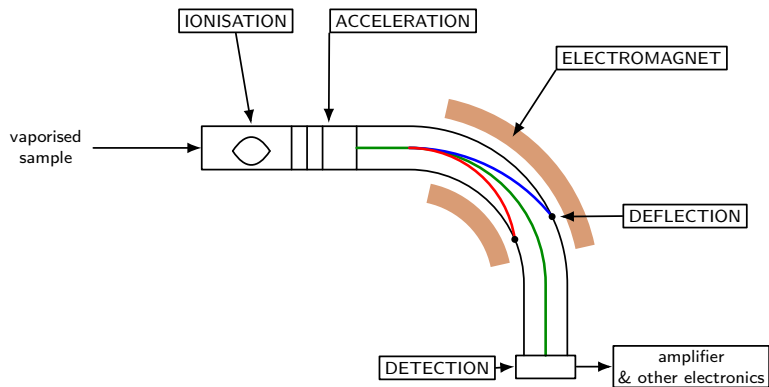
But why bother modelling this at all? What's the motivation?

Mass spectrometry is one of the most widely used tools we have for analysing molecules. But reading that data is still mostly done by hand, by experts, and it takes time.

So we can acquire spectra very fast — and none of that speed carries over into the analysis. That's where the bottleneck sits.

And the hope is: if we feed chemical structure into the models, they get more interpretable and they generalise better.

Mass Spectrometer



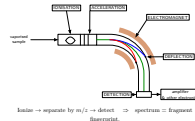
Ionize $\rightarrow$ separate by m/z $\rightarrow$ detect $\Rightarrow$ spectrum = fragment fingerprint.

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└ Background

└ Mass Spectrometer

Mass Spectrometer

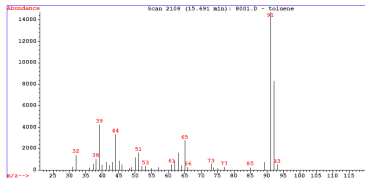
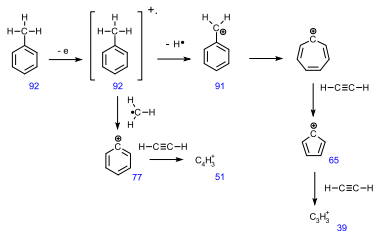


A quick reminder of how a mass spectrometer works. We start with a sample, and that sample gets ionised. The ions are then separated by their mass-to-charge ratio, and at the end they hit a detector — and that's what gives us the spectrum.

The part that matters for us: under electron ionisation the molecule doesn't stay in one piece, it breaks apart. So the spectrum is really a fingerprint of those fragments. And that's exactly what I exploit later on.

Data Sources

- Mass spectrometry data from NIST → SMILES strings & spectra
- Curated EI fragmentation reactions → MØD rules

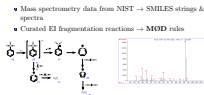


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└ Data

└ Data Sources

Data Sources



I work with two data sources.

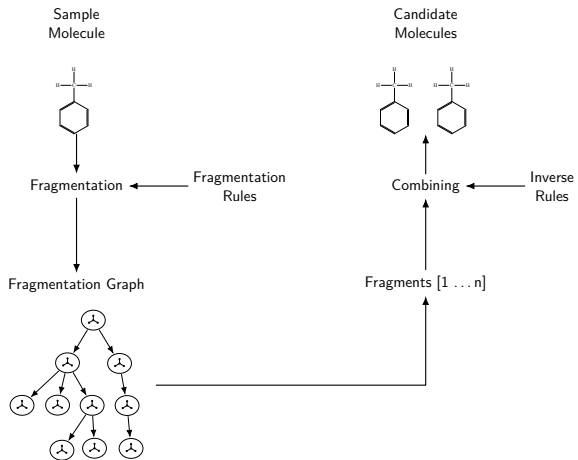
The first is mass-spectrometry data from NIST.

That gives me the structures, as SMILES strings, together with the measured spectra.

The second is curated EI fragmentation reactions from the literature, and those I encode as MØD rules.

On the left you see toluene falling apart, and on the right the spectrum that comes out of it.

Data Generation Overview

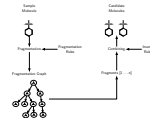


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└ Data

└ Data Generation Overview

Data Generation Overview



This is the whole pipeline in one picture.

I take a molecule and fragment it with the rules in MØD.

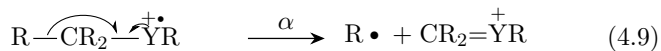
That gives me a fragmentation graph,

and from that graph I read off the fragments.

Then I go the other way round: I take those fragments and put them back together, using the inverse of the reaction rules.

And that's how I get candidate structures for a given spectrum.

Sample Fragmentation Rules



```

1  IMS_4_9 = mod.Rule.fromDFS(
2      s =
3      "[C]1[C]2([C]3)([C]4)[_A+.]5"
4      ">>"
5      "[C.]1" "." "[C]2([C]3)([C]4){=}[_A+]5",
6      name =
7      "radical induced (alpha-)cleavage for a saturated site"
8      " 4.9"
9      " $R1R3R4Y5"
10 )

```

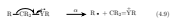
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└ Data

└ Sample Fragmentation Rules

Sample Fragmentation Rules



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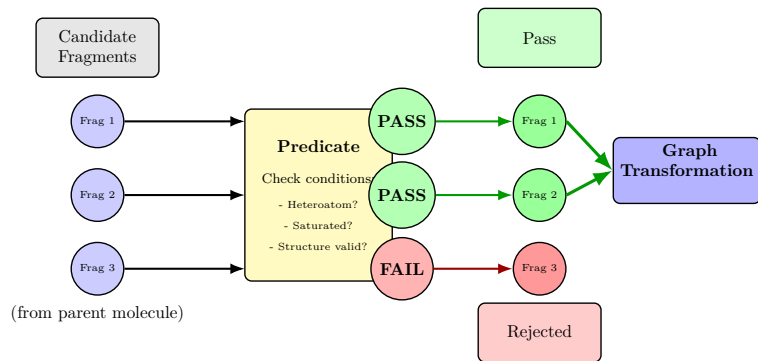
At the top you see one of these rules the way it's written in the literature: equation 4.9, radical-induced alpha-cleavage at a saturated site. The radical cation sits on the heteroatom Y, a bond to the neighbouring carbon breaks, and a new pi bond forms between that carbon and Y.

And the thing to notice is that the book doesn't only talk about single atoms — it also refers to whole groups. Those are the R and Y placeholders.

At the bottom is the same rule in code. The graph transformation itself only sees single atoms, so I carry the group conditions in the rule name, after that paragraph sign: here atoms 1, 3 and 4 have to be alkyl groups and atom 5 a heteroatom.

And that extended notation is what the predicates on the next slides enforce during the transformation.

MØD - Predicate



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└ Data

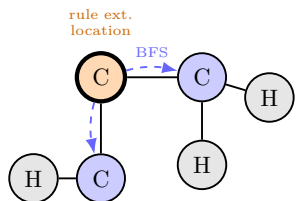
└ MØD - Predicate

MØD - Predicate



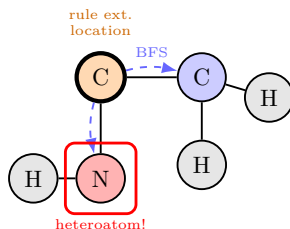
MØD lets you hook in so-called predicates. They're callbacks that get called during the transformation, and I use them to filter candidates according to the rule extension. On the next few slides I'll walk you through the ones I implemented.

MØD - Predicate - Alkyl Group



VALID: Alkyl Group

BFS neighbors $\subseteq \{H, C\}$ ✓



INVALID: Not Alkyl

BFS neighbors $\not\subseteq \{H, C\}$ ×

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└ Data

└ MØD - Predicate - Alkyl Group

MØD - Predicate - Alkyl Group

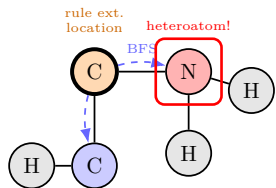


The first one is the alkyl group. An alkyl substituent contains nothing but carbon and hydrogen.

So I start where the rule is extended, and collect the neighbours breadth-first, until I run into atoms that are already part of the match of the original rule.

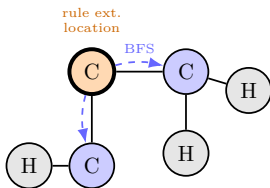
And then I simply check: is everything I collected a carbon or a hydrogen? If yes, it's a valid alkyl group. If not, I reject it.

MØD - Predicate - Heteroatom Group



VALID: Heteroatom Group

$$\text{BFS} \setminus \{\text{H}, \text{C}\} \neq \emptyset \checkmark$$



INVALID: No Heteroatom

$$\text{BFS} \setminus \{\text{H}, \text{C}\} = \emptyset \times$$

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└ Data

└ MØD - Predicate - Heteroatom Group

MØD - Predicate - Heteroatom Group



The heteroatom group is basically the mirror image of that. Here I need at least one atom that is *not* a carbon or a hydrogen.

Again I collect the neighbours breadth-first — the ones the original rule hasn't matched —

and then I check whether that set contains anything outside hydrogen and carbon.

If it does, it's a valid heteroatom group. If it's only hydrogen and carbon, it gets rejected.

MØD - Predicate - Saturation

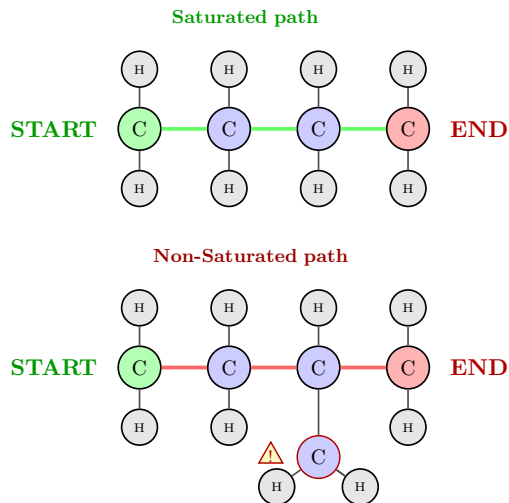
Goal: Find saturated path

Requirements:

- ✓ Only carbon backbone
- ✓ Single bonds only (C—C)
- ✓ Only H side branches

Method:

Search from **start** to **end**
following C-C single bonds



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└ Data

└ MØD - Predicate - Saturation

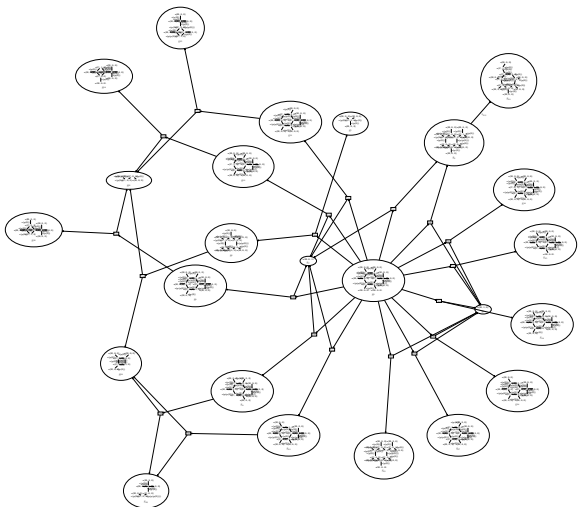
MØD - Predicate - Saturation



And the third one is saturation. Here I'm looking for a chain of carbons that are connected only by single bonds, and where every carbon carries nothing but hydrogens on the side.

So I search from the start atom to the end atom, along carbon-carbon single bonds. The example at the top passes. The one at the bottom fails, and there are two reasons for that: the branching carbon, and the double bond.

Sample Derivation Graph



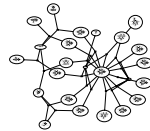
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└ Data

└ Sample Derivation Graph

Sample Derivation Graph

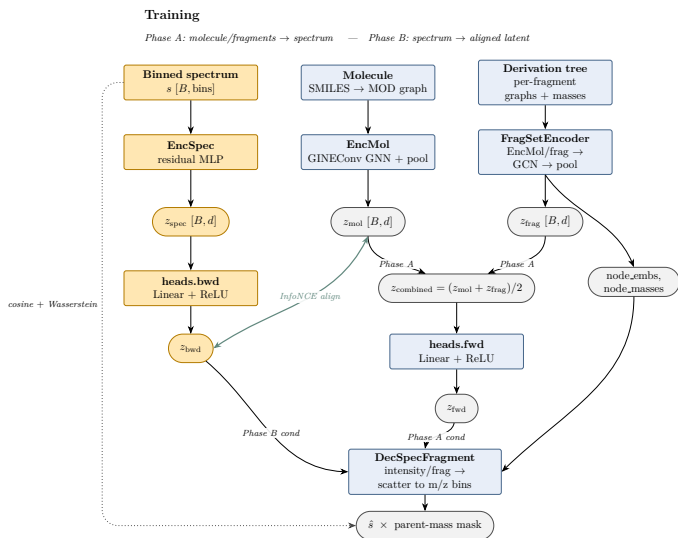


And this is what actually comes out of the fragmentation. It's drawn as a derivation graph.

The nodes are the fragments, and the edges are the reactions that connect them.

You're not meant to read every single node here. The point is just this: from one input molecule, MØD builds up a whole rich, connected space of fragments — and that's the space I then mine for candidate structures.

Architecture – Training



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└ ML-Modelling

└ Architecture – Training

Architecture – Training



This is the model I implemented. It takes three inputs: the molecule graph, the derivation graph — so the fragmentation — and the mass spectrum.

Each of them gets its own encoder, and they all map into one shared latent space. For the molecule that's a GCN with global pooling, for the derivation graph a two-level graph encoder, and for the spectrum a residual MLP.

Going forward, I fuse the molecule and the fragment latents, and decode them into a predicted spectrum.

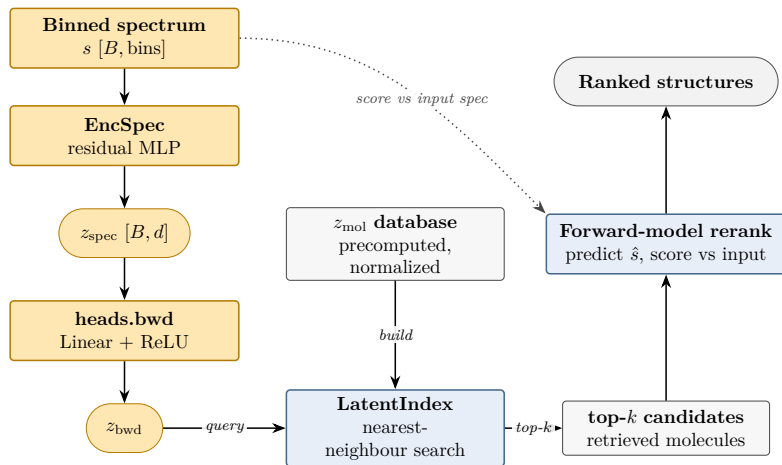
The backward direction deserves a comment, because it is *not* a spectrum-to-graph decoder. It's a retrieval path. I align the spectrum embedding with the molecular embeddings, so at test time I can pull out the nearest candidate structures by cosine similarity, and then rerank them with the forward model.

Both directions share the task heads and the decoder, and they're tied together by two auxiliary losses: cycle consistency, and an InfoNCE

Architecture – Inference

Inference / retrieval

Query with z_{bwd} into the molecule-latent index, then rerank with the forward model

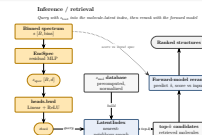


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└ ML-Modelling

└ Architecture – Inference

Architecture – Inference



And this is what inference looks like. I take a spectrum, and I embed it into the latent space.

Then I retrieve the nearest molecular candidates by cosine similarity, and rerank them with the forward model.

One caveat I want to put on the record here, because it comes up in the results: the reranking step does improve the raw retrieval numbers, but when I audited it, that improvement is the precursor-mass consistency filter inside it — swapping the predicted spectrum for a constant one reproduces the same gain. So I do not claim the learned spectral shape is what drives retrieval.

Results – Mass Sets the Bar

Held-out test split · $N = 13$ queries · gallery of 66 EI compounds

Ranking signal	MRR
True parent mass (oracle)	0.923
Spectrum-implied parent mass	0.462
Mass mask + cosine	0.498
Learned embedding	0.112

⇒ uncontrolled retrieval scores measure **mass agreement**, not chemistry
 — so they are not the axis.

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 ↳ Results

↳ Results – Mass Sets the Bar

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Now to the results, and I want to start with the thing that decides how every other number here has to be read.

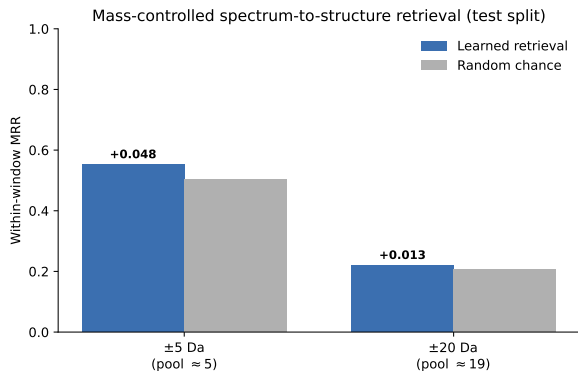
In this retrieval setting, parent mass alone is almost a solution. An oracle that ranks candidates by their true mass reaches a mean reciprocal rank of 0.92. Even a mass estimate read straight off the spectrum’s heaviest peak gets 0.46.

My learned embedding, scored on the full set, gets 0.11.

So any raw retrieval score I could put on a slide sits under a very strong mass baseline — including the apparent benefit of forward-model reranking, which is really the mass-consistency mask doing the work: mask plus cosine lands at 0.50, right at the mass baseline.

That is why the honest question is not ”how high is the retrieval score”, but ”is there anything left once mass is controlled for”. Which is the next slide.

Results – Mass-Controlled Retrieval



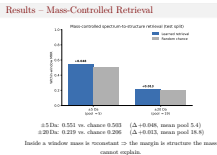
±5 Da: 0.551 vs. chance 0.503 ($\Delta +0.048$, mean pool 5.4)
 ±20 Da: 0.219 vs. chance 0.206 ($\Delta +0.013$, mean pool 18.8)

Inside a window mass is $\approx$ constant $\Rightarrow$ the margin is structure the mass cannot explain.

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 └ Results

└ Results – Mass-Controlled Retrieval



Here mass is held fixed. I take a window around the query's true mass — plus/minus 5 Dalton, and plus/minus 20 — and rank only the candidates inside it. Within such a window every candidate has essentially the same mass, so mass can no longer do the ranking for me.

Against the analytic random-chance level for a pool of that size, the learned retrieval comes out at 0.551 versus 0.503 in the tight window, and 0.219 versus 0.206 in the wide one. So a margin of about five and one percentage points.

These are small margins, on thirteen held-out queries, from a single training run — I am not going to oversell them. But they are positive on molecules that were never seen in training or model selection, and they are the only part of the evaluation that mass cannot account for.

Conclusions

Contributions

- **Fragment-grounded forward model:** intensities scattered from the molecule's own MØD fragments $\Rightarrow$ molecule-specific *by construction*
- **Mass-controlled evaluation:** separates learned structure from parent-mass filtering
- Bidirectional framework ($f : M \rightarrow S, f^{-1} : S \rightarrow M$) + rule-extension DSL

Honest assessment

- Beyond-mass margin is *small*: +0.048 / +0.013 MRR on 13 queries, single run
- Uncontrolled metrics — reranking included — are mass filtering, not evidence
- Fragment branch buys molecule-specificity, not a retrieval win

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└ Conclusions

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Honest assessment

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So, to wrap up. Two contributions I would defend.

First, the forward model is grounded in fragments: every predicted intensity is scattered from the molecule's own MØD fragments, so it is molecule-specific by construction — a design property, not a lucky result.

Second, and more important: the evaluation. Mass is such a strong signal here that most retrieval numbers are mass filtering in disguise. The mass-controlled protocol separates the two.

And around those, the bidirectional framework and the rule-extension DSL.

Honestly: what survives the control is small — plus five and plus one points of MRR, thirteen queries, one run. The uncontrolled metrics, reranking included, are mass consistency and I don't present them as evidence. The fragment branch earns its place through molecule-specificity, not a retrieval win. Which is the setup for the outlook.

Outlook – Learn to *Generate* Fragmentation

Model *encodes* precomputed MØD graphs — it does *not* learn fragmentation.

Inspiration: **ICEBERG** [4] — *Generate–Score*.

- Have: the **Score** side
- Missing: a **Generate** module (which fragmentations?)
- MØD graphs → proxy supervision

prune/rank → autoregressive expansion → joint end-to-end

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└─ Conclusions

└─ Outlook – Learn to *Generate* Fragmentation

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Inspiration: **ICEBERG** [4] — *Generate–Score*.

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prune/rank → autoregressive expansion → joint end-to-end

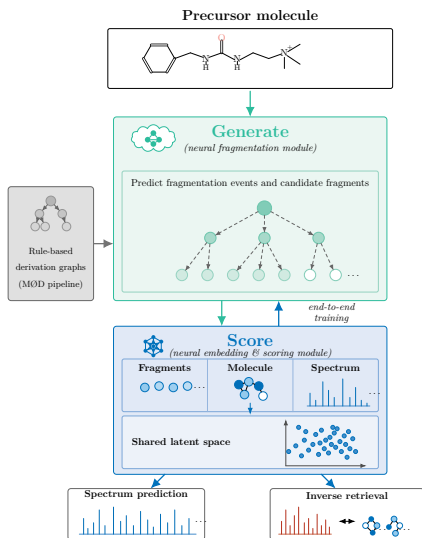
The clearest limit of what I’ve shown you is this: the model doesn’t learn fragmentation. It encodes derivation graphs that the rule-based MØD pipeline has already computed upstream. The learning problem only starts once the fragmentation space is built.

So the natural next step — and that’s where I’d like to take this — is to move from encoding fragmentation to generating it. The main inspiration here is ICEBERG, by Goldman and colleagues. They split the task in two: a Generate module that proposes likely fragmentation events, and a Score module that assigns the intensities.

And if you look at my model through that lens, I already have the Score side. It embeds molecules, fragments and spectra into a shared space, it predicts spectra, it aligns and retrieves. What’s missing is the Generate module upstream — the part that decides which fragments even enter the picture. And the symbolic MØD derivation graphs could serve as proxy supervision for training exactly that.

I’d do it in stages: first learn to prune or rank derivation steps, then

Outlook – Toward a Generate-Score Architecture



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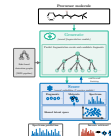
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 └ Conclusions

└ Outlook – Toward a Generate-Score Architecture

And this is roughly what that would look like.

A learned Generate module, trained with the MØD derivation graphs as proxy labels, proposes the fragmentation structure. And the Score module — that's essentially my current model — embeds the molecule, the fragments and the spectrum into a shared latent space, and produces both the forward spectrum prediction and the inverse retrieval. The two would be trained end-to-end. So the model wouldn't just learn how to *use* fragments — it would learn which fragmentation structures to build in the first place.

Outlook – Toward a Generate-Score Architecture



References

[1] V8rik. *Toluene Fragmentation*. 2008. URL: <https://upload.wikimedia.org/wikipedia/commons/6/6c/TolueneFragmentation.svg> (visited on 01/20/2026).

[2] Paginazero, Ronhjones. *Toluene Spectra*. 2017. URL: https://upload.wikimedia.org/wikipedia/commons/2/2c/Spettro_di_massa_del_toluene.PNG (visited on 01/20/2026).

[3] Fred W McLafferty and František Tureček. *Interpretation von Massenspektren*. ger. Spektrum Lehrbuch. Berlin [u.a.]: Spektrum Akad. Verl., 1995. ISBN: 9783642398483.

[4] Samuel Goldman, Janet Li, and Connor W. Coley. “Generating Molecular Fragmentation Graphs with Autoregressive Neural Networks”. In: *Analytical Chemistry* 96.8 (Feb. 2024), pp. 3419–3428. ISSN: 1520-6882. DOI: 10.1021/acs.analchem.3c04654.

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└─Conclusions

└─References

And these are the sources and references I used.

References

[1] V8rik. *Toluene Fragmentation*. 2008. URL: <https://upload.wikimedia.org/wikipedia/commons/6/6c/TolueneFragmentation.svg> (visited on 01/20/2026).

[2] Paginazero, Ronhjones. *Toluene Spectra*. 2017. URL: https://upload.wikimedia.org/wikipedia/commons/2/2c/Spettro_di_massa_del_toluene.PNG (visited on 01/20/2026).

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Problem	Background	Data	ML-Modelling	Results	Conclusions
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Discussion					

Questions?

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└─ Conclusions
 └─ Discussion

Discussion

Questions?

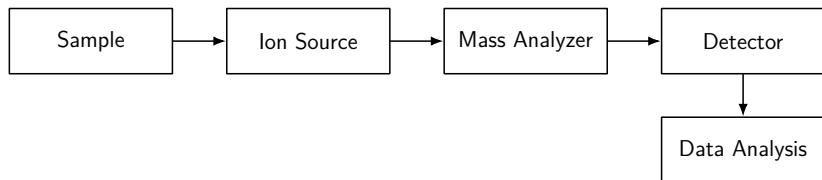
That's it from my side — thank you for your attention.
I'm happy to take any questions.

Supplementary Figures

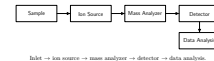
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- ◆ Data-generation cost

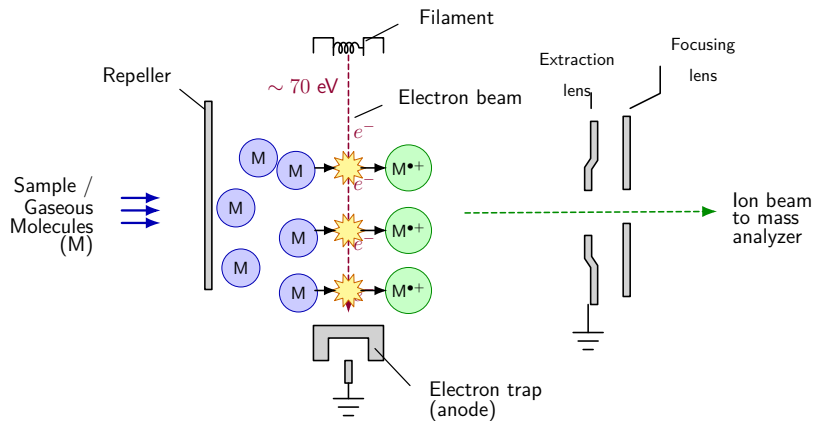
These are backup slides — the figures from the thesis that didn't make it into the talk. I'll jump to whichever one is relevant to your question.



Inlet $\rightarrow$ ion source $\rightarrow$ mass analyzer $\rightarrow$ detector $\rightarrow$ data analysis.



Electron Ionization (EI)



~ 70 eV electrons eject one electron $\Rightarrow$ radical cation $[M]^{\bullet+}$, which then fragments.

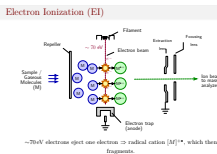
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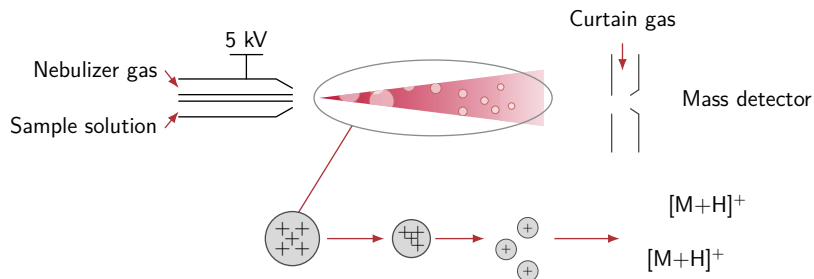
-Instrumentation & Ionization

- Electron Ionization (EI)



This is the ionization method this whole thesis is about. A 70-electron-volt beam knocks out one electron, you get a radical cation, and its excess internal energy is what drives the fragmentation.

Electrospray Ionization (ESI)



Soft ionization: charged droplets $\rightarrow$ gas-phase ions, typically $[M+H]^+$.

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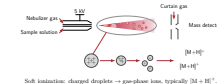
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Instrumentation & Ionization

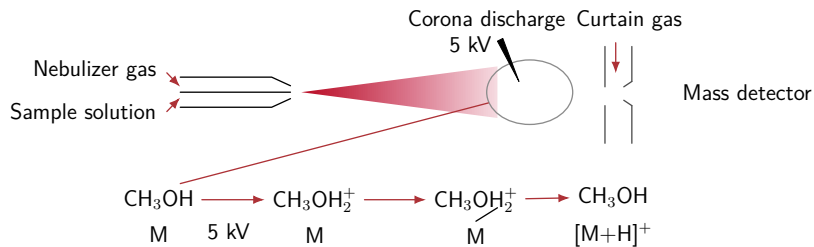
Electrospray Ionization (ESI)

Electrospray Ionization (ESI)



ESI for contrast: a soft method, so you mostly keep the intact molecule and get very little fragmentation — which is exactly the information my approach relies on.

Atmospheric Pressure Chemical Ionization (APCI)



Corona discharge makes reagent ions; proton transfer gives [M + H]⁺.

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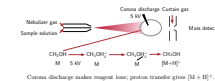
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Instrumentation & Ionization

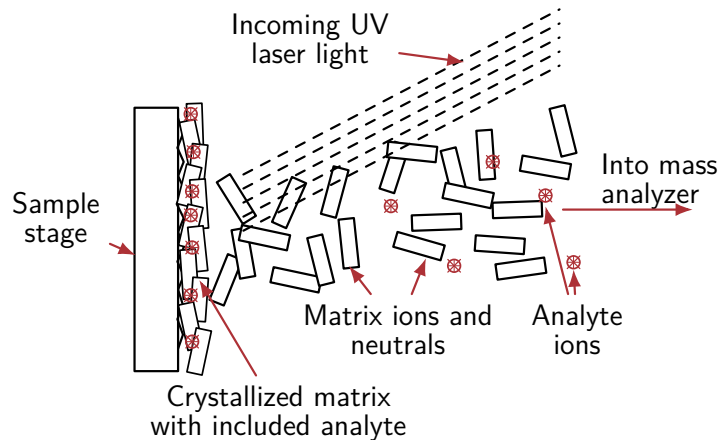
Atmospheric Pressure Chemical Ionization (APCI)

Atmospheric Pressure Chemical Ionization (APCI)



APCI: the solvent is ionized first by a corona discharge, and those reagent ions then hand a proton to the analyte.

Matrix-Assisted Laser Desorption/Ionization (MALDI)



Laser desorption from a matrix; predominantly singly charged ions, suited to large biomolecules.

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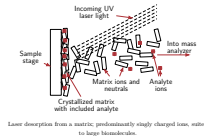
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└ Instrumentation & Ionization

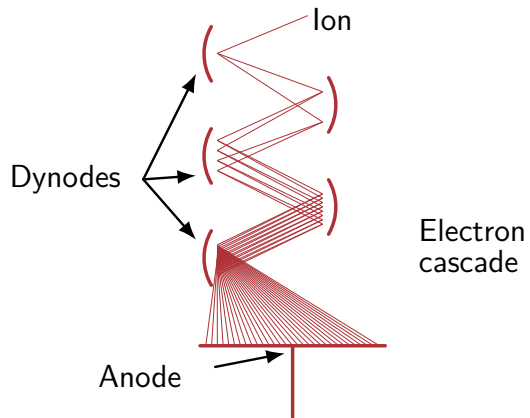
└ Matrix-Assisted Laser Desorption/Ionization (MALDI)

Matrix-Assisted Laser Desorption/Ionization (MALDI)



MALDI: the analyte sits embedded in a matrix, a laser pulse desorbs both, and the matrix ions do the ionizing. Mostly singly charged, and it scales to large biomolecules.

Electron Multiplier Detector



Cascading secondary electrons across dynodes amplify a single ion into a measurable current pulse.

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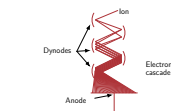
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└ Electron Multiplier Detector

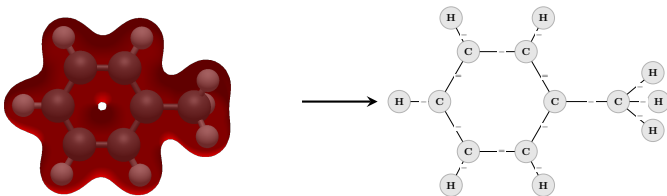
Electron Multiplier Detector



Cascading secondary electrons across dynodes amplify a single ion into a measurable current pulse.

And the detection side: one ion hits the first dynode, each stage multiplies the electrons, and after several stages you have a current pulse you can actually digitize.

Molecule as a Graph



Geometry and electron density are abstracted away; connectivity is kept.

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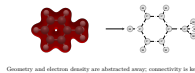
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└ Molecular Representation

└ Molecule as a Graph

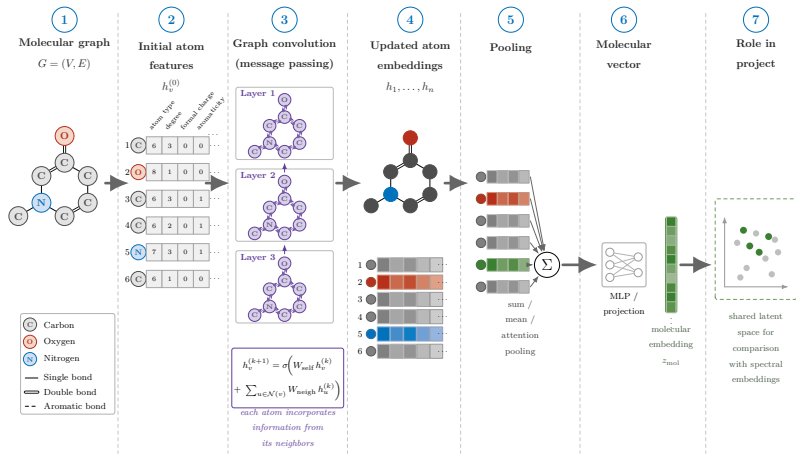
Molecule as a Graph



Geometry and electron density are abstracted away; connectivity is kept.

The abstraction everything rests on: toluene as a quantum-mechanical object on the left, as a graph on the right. Atoms become vertices, bonds become edges, and geometry and electron density are dropped.

Graph-Convolutional Molecular Encoding



Atom features $\rightarrow$ message passing $\rightarrow$ pooling $\rightarrow z_{mol}$ in the shared latent space.

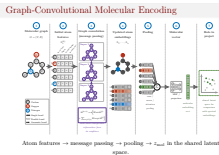
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Molecular Representation

Graph-Convolutional Molecular Encoding



How the molecule encoder works in detail. Each atom starts with a feature vector, message passing makes those embeddings context-aware, pooling collapses them into one fixed-length vector, and a projection puts it into the shared latent space.

Two-Level Representation – Overview



Local graph encoders for atom/bond chemistry, global hypergraph encoders for multi-step fragmentation.

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└ Molecular Representation

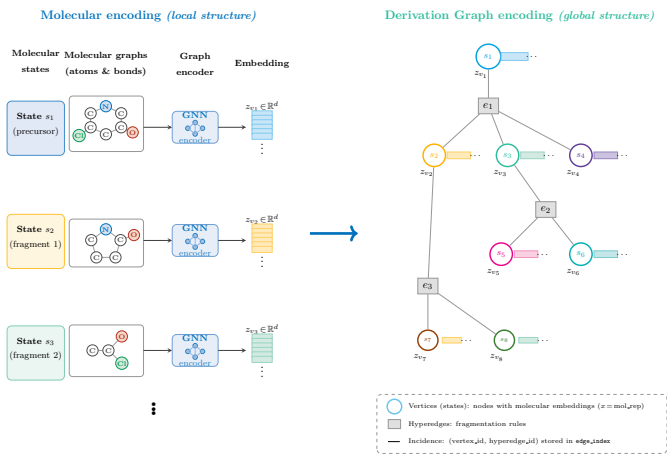
└ Two-Level Representation – Overview

Two-Level Representation – Overview

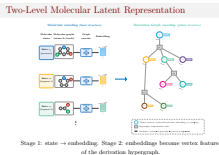


The two-level idea in one picture: the lower level encodes the chemistry inside each molecular state, the upper level encodes how those states are connected by fragmentation.

Two-Level Molecular Latent Representation



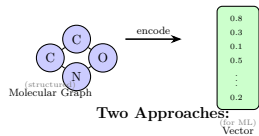
Stage 1: state $\rightarrow$ embedding. Stage 2: embeddings become vertex features of the derivation hypergraph.



And the same thing spelled out concretely. First every molecular state is encoded on its own. Then those embeddings go in as vertex features of the derivation hypergraph, so the hyperedges carry the fragmentation structure on top.

The Graph Representation Challenge

Graph Representation Challenge



Two Approaches:

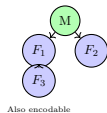


✓ Better performance
✗ Not invertible



✓ Structured info
✓ For fragments too

Fragmentation Trees



Also encodable

Fingerprints score well but are not (easily) invertible — which matters for the inverse direction.

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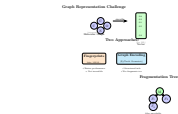
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└ Molecular Representation

└ The Graph Representation Challenge

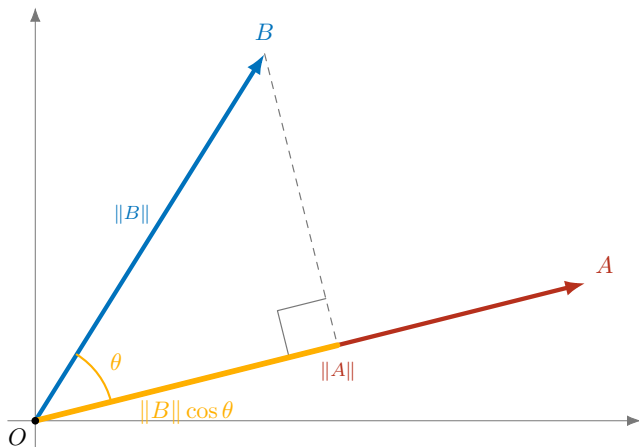
The Graph Representation Challenge



Fingerprints score well but are not (easily) invertible — which matters for the inverse direction.

Why the representation question is not settled: fingerprints perform very well, but you cannot easily invert them back to a structure. And the inverse direction is exactly what I need.

Cosine Similarity – Geometry



$$\cos \theta = (A \cdot B) / (\|A\| \|B\|) \text{ — direction only, hence scale-invariant.}$$

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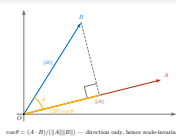
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└ Spectral Comparison Measures

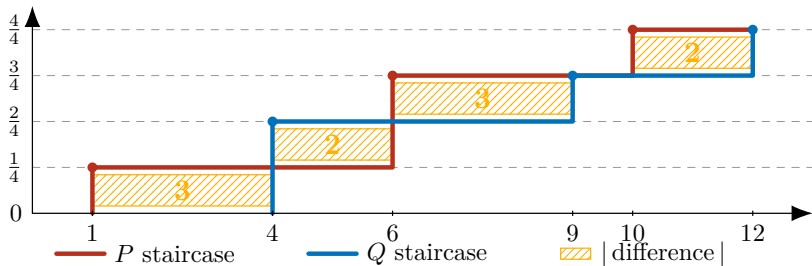
└ Cosine Similarity – Geometry

Cosine Similarity – Geometry


 $\cos \theta = (A \cdot B) / (\|A\| \|B\|)$ — direction only, hence scale-invariant.

Cosine similarity is just the angle between the two intensity vectors. It depends on direction only, so rescaling a spectrum doesn't change it.

Wasserstein Distance – Illustration



Sorted vectors as cumulative staircases; the distance is the shaded transport area, normalised by n .

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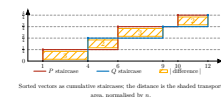
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Spectral Comparison Measures

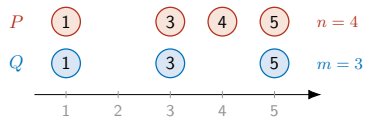
Wasserstein Distance – Illustration

Wasserstein Distance – Illustration



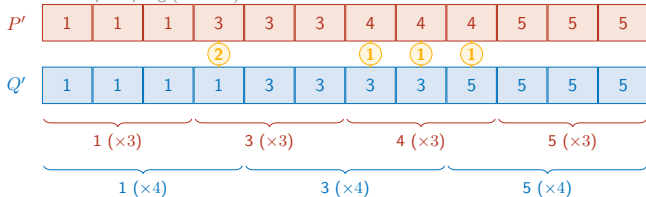
The Wasserstein distance as transport cost: draw both sorted vectors as cumulative staircases, and the shaded area between them — divided by the number of elements — is the distance.

Wasserstein Distance – Upsampling



LCM(4, 3) = 12 — upsample to equal length
 repeat each P element $\times 3$ · repeat each Q element $\times 4$

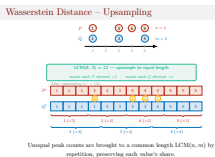
After upsampling ($n' = 12$):



Unequal peak counts are brought to a common length $\text{LCM}(n, m)$ by repetition, preserving each value's share.

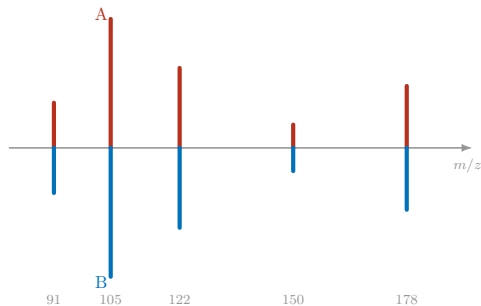
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 └ Wasserstein Distance – Upsampling



The matched-pair form needs both vectors to have the same length, and real spectra never do. So each is repeated up to the least common multiple, which keeps every value's relative share of the total mass.

Cosine vs. Wasserstein – (a) Aligned Spectra

(a) Aligned spectra — identical, $\theta = 0$ 

cosine = 1.00 Wasserstein = 0.0

Every B peak meets its A counterpart: cosine = 1 ($\theta = 0$), Wasserstein = 0
(no mass to move).

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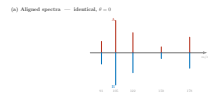
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└ Cosine vs. Wasserstein – (a) Aligned Spectra

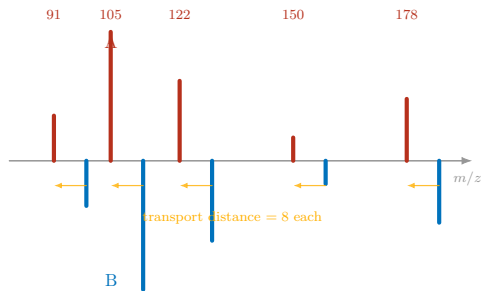
Cosine vs. Wasserstein – (a) Aligned Spectra



cosine = 1.00 Wasserstein = 0.0
Every B peak meets its A counterpart: cosine = 1 ($\theta = 0$), Wasserstein = 0
(no mass to move).

The reference case: spectrum A upward, spectrum B mirrored downward, peaks aligned. Both measures agree that these are identical.

Cosine vs. Wasserstein – (b) B Shifted +8 Da

(b) B shifted +8 Da — peaks miss, $\theta \rightarrow 90^\circ$ cosine = **0.00** (no shared peaks) Wasserstein = **21.1**

Cosine counts only *shared* positions $\Rightarrow$ collapses to 0; Wasserstein transports the mass back and stays finite (21.1).

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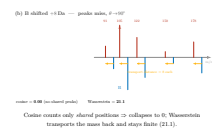
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Spectral Comparison Measures

Cosine vs. Wasserstein – (b) B Shifted +8 Da

Cosine vs. Wasserstein – (b) B Shifted +8 Da



Now shift B by 8 Dalton. Cosine only sees intensity at shared positions, so its dot product vanishes and it drops straight to zero — even though the spectra are obviously still related. Wasserstein measures the work of moving B's mass back onto A and returns a finite number.

Cosine vs. Wasserstein – (c) Response vs. Shift

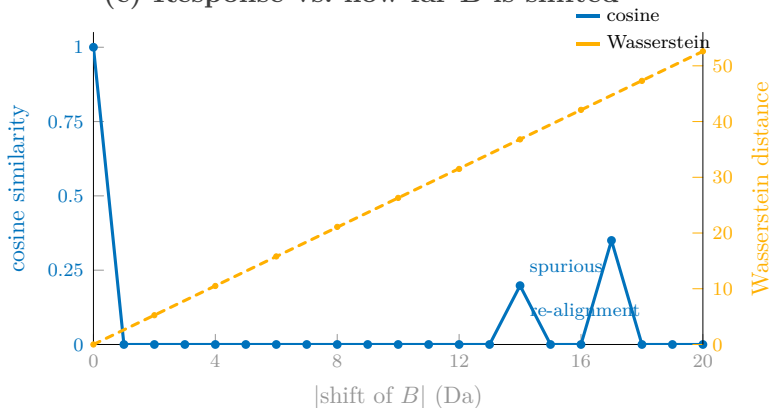
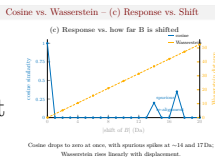
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Spectral Comparison Measures

Cosine vs. Wasserstein – (c) Response vs. Shift



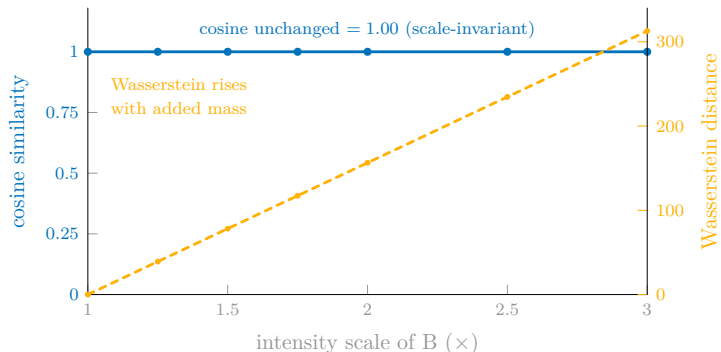
Cosine drops to zero at once, with spurious spikes at ~ 14 and 17 Da;

Wasserstein rises linearly with displacement.

Sweeping the shift makes it systematic. Cosine falls off a cliff and even shows spurious spikes, where a displaced peak accidentally lands on a different peak. Wasserstein degrades gracefully, in proportion to the displacement.

Cosine vs. Wasserstein – (d) Response vs. Scaling

(d) Response vs. B's peak intensity (positions aligned)



Cosine unchanged at 1.00 (direction, not magnitude); un-normalized
Wasserstein grows with the added intensity.

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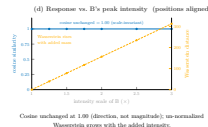
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Spectral Comparison Measures

Cosine vs. Wasserstein – (d) Response vs.
Scaling

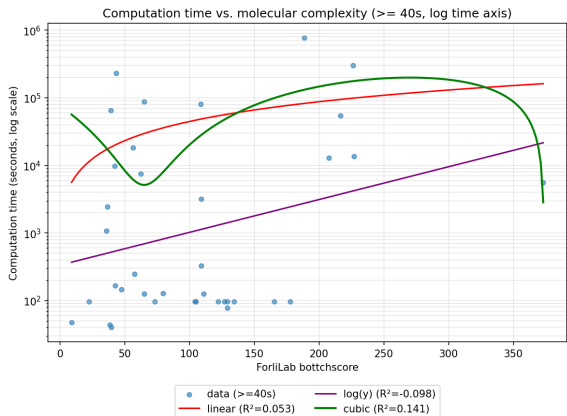
Cosine vs. Wasserstein – (d) Response vs. Scaling



And the complementary axis: scale B's intensities with the positions held fixed. Cosine doesn't move, because it compares direction, not magnitude. The un-normalized Wasserstein distance grows with the added mass.

So the two measures capture different things — pattern at fixed coordinates versus position along the axis — which is why the loss uses both.

Fragmentation Computation Times



Time vs. molecular complexity (trivial expansions < 40 s excluded):
positive trend, but huge variance at equal complexity.

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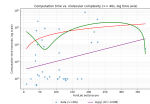
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└ Fragmentation Computation Times

Fragmentation Computation Times



Time vs. molecular complexity (trivial expansions < 40 s excluded):
positive trend, but huge variance at equal complexity.

Cost against molecular complexity, on a log scale, with the trivial expansions cut off. There is a clear positive trend, but the spread at any given complexity is enormous — so complexity alone doesn't predict runtime; the available fragmentation pathways matter just as much.

Where the Data-Generation Time Goes

MØD data generation — sampled call tree (folic_acid, forward-only, 1 thread)

width x wall time - 11,662 stack samples over 117.9 s of build (imports pre-warmed) - 86% of samples are inside MØD's C++ matching loop with no Python frame below



Sampled call tree (folic acid, forward-only, 1 thread): 86% of samples sit inside MØD's C++ matching loop.

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└ Where the Data-Generation Time Goes

Where the Data-Generation Time Goes



Sampled call tree (folic acid, forward-only, 1 thread): 86% of samples sit inside MØD's C++ matching loop.

And a profile of where that time actually goes. The vast majority of samples are inside MØD's C++ matching loop, with no Python frame below — so the Python-side predicates are not the bottleneck, and speeding this up is not a matter of optimizing my own code.

Splits, Gallery and Provenance

Molecules with paired dumps + spectra	66
train / validation / test	38 / 15 / 13
Retrieval gallery (train + val + test)	66
Mass-controlled queries (test)	13
Training runs behind the numbers	1
Spectrum bins (1–1000 Da, 1 Da)	999

All reported numbers come from one training run (`ml_checkpoint.pt`, 2026-07-06) — no seed averaging, no model selection on the test split.

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└ Splits, Gallery and Provenance

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All reported numbers come from one training run (`ml_checkpoint.pt`, 2026-07-06) — no seed averaging, no model selection on the test split.

The numbers behind the results slides. 66 molecules had both a derivation graph and a measured spectrum; that splits 38 / 15 / 13. Retrieval is closed-library: the gallery is all 66, which is the structure-elucidation setting, and the 13 test molecules were never seen in training or in model selection.

Everything comes from a single training run. No seed averaging. That is the main reason I keep saying proof of concept.

Training Hyperparameters

Optimiser	Adam (one per phase)
Learning rate / weight decay	7.1×10^{-5} / 1.4×10^{-4}
Batch size	64
Epochs (Phase A + Phase B)	10 000 + 10 000
Shared latent dimension d	256
GINEConv layers / width	3 / 128
Derivation-graph GCN	2× GCNConv, mean pool
Dropout / normalisation	0.43 / BatchNorm
Cosine / Wasserstein weight	1.59 / 1.17
Cycle (fwd / spec) weight	0.77 / 0.089
Align weight / temperature τ	0.30 / 0.07
Curriculum start / increment	0.73 / 0.17

Tuned entries: best of a 20-trial Optuna sweep; checkpoint selection = final epoch.

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Tuned entries: best of a 20-trial Optuna sweep; checkpoint selection = final epoch.

And the hyperparameters, if anyone wants them. The tuned ones are the best trial of a twenty-trial Optuna search — learning rate, weight decay, batch size, latent dimension, dropout, normalisation, the loss weights and the curriculum. The rest were fixed a priori.